Poonawalla Fincorp Limited

Cloud Adoption Policy

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CLOUD ADOPTION POLICY

Document review and approval

Document Control

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Date	**Version**	**Change Description (in brief)**
20th - OCT-23	1.0	New policy
25th -OCT- 24	2.0	Change in Policy Owner
17th -OCT-25	2.0	Annual Review- No changes suggested

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1. Overview

This policy confirms the cloud adoption by ensuring confidentiality, integrity and availability of data stored, accessed and classification using cloud computing services. This policy provides due diligence to manage and continually monitor the risk associated with system deployed in cloud service provider's environment.

2. Objective

All services within the cloud environment will be subject to the requirements specified within this policy. Therefore, it applies to every component server, database, and other IT system in cloud computing environment. It establishes a framework to meet regulatory requirements and cloud computing guidelines.

3. Scope

The cloud adoption policy is defined to oversee, control, secure and to manage all types of cloud services managed by PFL. This policy shall be applicable to all types of cloud services, cloud models managed by PFL.

4. Ownership and Responsibilities

4.1 Information Technology Team

1. Evaluate and select the cloud platform based on the business requirement.
2. Select appropriate cloud computing service model.
3. Provide technical expertise and support.
4. Hosting applications and infrastructure on cloud ecosystem
5. Implement security measures defined by information security team.

4.2 Information Security Team

1. Design security controls and processes
2. Conduct risk assessments on PFL managed cloud services
3. Conduct security reviews on PFL managed cloud services

4.3 Cloud Services Provider

1. Ensure compliance with the agreed requirements and local regulations.
2. Ensure availability and continuity of services as agreed.
3. Ensure that data stored on the cloud systems are within territorial jurisdiction of India.

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5. Cloud Computing Services

i. Software as a Service (SaaS).

Client uses the application(s) provided by the service provider on a cloud infrastructure.

ii. Platform as a Service (PaaS).

The service provides software for building applications, middleware, database, development environment and other tools along with the infrastructure to the client.

iii. Infrastructure as a Service (IaaS).

The service provides compute, storage, network, and other basic resources so that the client can develop and deploy their applications.

6. Cloud Deployment Models

i. Private cloud.

The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.

ii. Community cloud.

The cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.

iii. Public cloud.

The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.

iv. Hybrid cloud.

The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

7. Policy Statement

The Information Technology service delivery model of the PFL includes Cloud to deliver the infrastructure, platform or services as per the business needs. The PFL shall use the cloud in a variety of deployment models such as private cloud, public cloud or hybrid or multi cloud, and delivery models such as IaaS, PaaS and SaaS as may be deemed appropriate based upon the cost benefit,

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technology benefit, data security (as per the Information Security Policy), other risks and adherence to the guidelines of the PFL and the regulator.

8. Guiding Principles

Cloud First: For any new technology or business initiative, PFL shall consider cloud as the first option by selecting the most suitable combination of private or public or hybrid or multi cloud or on-premises infrastructure; and IaaS, PaaS or SaaS.

9. Policy

1. Cloud Only; PFL shall maximize the adoption of cloud by continuously as application landscape for its feasibility to migrate to cloud with the required compliance, cost benefit analysis, analyzing the security aspects as per the Information Security policy, data security / privacy.
2. The PFL shall regularly assess and migrate application to the private cloud (of the PFL)
3. Buy before Build: For new or upgrade to technology services and/ or platform shall consider "Buy", preferably through SaaS in cloud terms, before selecting the option provided it also gives a cost benefit and comply with security standards of the (as per the Information Security Policy). Examples of such cases are HRMS, Management tools etc.
4. Application Development: All application developments and upgrades to be ready whether in-house or outsourced, PFL shall assess the requirements and architecture style such as multi-cloud architecture, cloud-native architecture, architectures, containers and micro-services.
5. Best-in-class IT Operations: PFL shall regularly perform an application modernization / rationalization exercise to use_ latest technologies such as virtualization, containerization etc. and cloud integration platforms in order to bring cloud like efficiency and functionality to on-premises infrastructure and private cloud. PFL shall preferably use a single control pane to maintain, manage and monitor all the cloud environments to ensure seamless and efficient service management & support operations.
6. Lift-and-Shift: For any application where modernization is not possible or is a Commercial-Of-The-Shelf (COTS) product or change to the application may cause potential disruptions in the business/ operations, then for such application PFL shall lift-and-shift to the cloud provided its data or its dependent/ interfacing applications had been already moved to the cloud Based on the case and secured portability of the application, PFL shall lift-and shift the application to the cloud
7. Selection and evaluation of cloud services which includes deployment model, services shall be done by Information technology team.

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8. Evaluation of cloud services shall be done as per security measures indicated by PFL information security team.
9. While evaluating cloud services for PFL, cloud management team shall ensure that the appropriate cost benefit analysis is done and recorded before finalizing.
10. While adopting cloud services, ensure that all local applicable regulatory, statutory compliances are adhered to and cloud service provider is compliant with the records
11. While selecting cloud service providers, ensure that the data to be stored on cloud service provider resides within the local boundaries of country.
12. PFL Information technology team shall ensure that sufficient monitoring controls are in place to monitor cloud service provider.
13. PFL Information technology team shall ensure that periodic review of availability of cloud resources and plan overall capacity of cloud resources.
14. PFL Information security team shall ensure that reviews are being conducted on measures deployed to ensure continuity of cloud services in terms of BCP and DR
15. While evaluating and selecting cloud services, the best practices below shall be adopted by PFL Information Technology team (below) but not limited to shall be considered.
    i. Application Design
    ii. Business Continuity and recoverability
    iii. Data Location/Storage
    iv. Service Level Agreements (SLA)
    v. Security
16. All the records while adopting any cloud services shall be preserved.
17. All policies and procedures in terms of managing IT general controls shall be enforced while adopting cloud services, these controls are but limited to,
    vi. Backup and recovery
    vii. Incident Management
    viii. Change Management
    ix. User access
18. All PFL managed cloud services shall be periodically reviewed and assessed by the Information security team.
19. The cloud management team shall ensure that MIS reports shall be prepared and updates are shared with CIO/CTO on a periodic basis.
20. All cloud environments shall be backed up adequately as per PFL backup management SOP.

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21. At the end of contract expiration or while disengaging exit procedure / agreement termination procedures As per PFL offboarding checklist/process.
22. The exit strategy and service level stipulations in the SLA shall factor in,
    i. agreed processes and turnaround times for returning the RE's service collaterals and data held by the CSP.
    ii. data completeness and portability.
    iii. secure purge of RE's information from the CSP's environment.
    iv. smooth transition of services; and
    v. unambiguous definition of liabilities, damages, penalties, and indemnities.
23. PFL shall ensure that data/ information stored in the cloud to comply with integrity including creation / collection, classification, usage, storage, retention, and disposition as per the guidelines of the PFL and / or regulator.
24. The PFL shall use the necessary security controls, data encryption and other security related controls for the data, as per Data/ Information Security Classification standard, before storing the data/ information in the cloud or adopting any cloud service.
25. The PFL shall ensure that the data/ information stored in the cloud resides within the legislative boundaries of India, along with the adequate security safeguards to prohibit unauthorized access.
26. The PFL shall periodically assess that the information and security standards are met by the cloud which includes information security, privacy, data residency, usage, retention and disposition, repatriation, availability business continuity planning, emergency response and accessibility during the period of validity of the contract and in the event of a planned or unplanned termination of the cloud services.
27. PFL shall ensure that data/ information residing on the cloud can be replicated/ backed-up and be accessible to the PFL in the event the Cloud becomes in accessible (e.g due to a disaster event or termination of service) as per the PFL's business continuity policy.
28. PFL shall develop a business continuity plan for the cloud and adequately test it.
29. Exit Guidelines,
    i. Define an approach to cxit from the cloud for any scenario such as the cloud service provider goes out of business, or poor service by the cloud service provider, or non-compliance to PFL/ legal/ regulatory guidelines, or the PFL decides to stop taking services from the cloud service provider.
    ii. The exit approach to ensure that the PFL can withdraw data/ application/ services from the cloud in a safe, efficient, and cost-effective manner.
    iii. The exit approach to address the various lock-in risks of cloud such as data lock-in, sensitive data on private cloud/ on-premises. The PFL shall use suitable obfuscation technology (for data masking, encryption) for sensitive data (as defined in the Information

Security Policy and Procedure) to be stored on cloud. The PFL shall, after detailed assessment, prefer this deployment model to host.

30. Data Lock-in,
    i. PFL shall maintain an inventory of all types of data and applications that are run and managed on the cloud platform.
    ii. PFL shall ensure that the data export from the cloud, when needed, is provided in a portable format such as raw SQL, XML, and JSON etc. that can be accepted by the on-premise tool.
    iii. PFL shall preferably use "The Open Data Element Framework" to standardize the documentation, categorization, indexing and extraction of data from the cloud.
    iv. PFL shall ensure that the data transfer does not lead to any loss of application functionality, data, performance, and information security.
31. Application Lock-in,
    i. The PFL shall avoid any specific components, technology or services of any particular cloud/which makes moving the application to another cloud or on-premise extremely difficult and/ or expensive.
    ii. The PFL shall always use standard interfaces and/or APIs; and avoid using any proprietary specifications and standards for integration of applications to the cloud.
    iii. The PFL shall ensure that applications that are to be migrated to cloud should be developed using a modular/ loosely-couple architecture with clear segregation of business logic and application logic.
32. Infrastructure Lock-In,
    i. The PFL shall use container technologies like Docker, Kubernetes, Open shift Cloud Foundry ete but not limited to. To isolate the application from its infrastructure so that the application is not dependent on a specific kind of hardware that may be provided by a particular cloud.
    ii. The PFL shall implement configuration and deployment management tools (DevOps tools such as Chef, Puppet) to automate the configuration and deployment to diverse environments making it easier to move from one cloud to another or to on-premise.
33. Information Management and Security,
    i. PFL shall ensure that data/ information stored in the cloud to comply with integrity including creation / collection, classification, usage, storage, retention, and disposition as per the guidelines of regulator, and Information Security Policy of the PFL.
    ii. The PFL shall use the necessary security controls, data encryption and other security related controls for the data, as per Data/ Information Security Classification standard, before storing the data/ information in the cloud or adopting any cloud service.

iii. The PFL shall ensure that the data/ information stored in the cloud resides within the legislative boundaries of India, along with the adequate security safeguards to prohibit unauthorized access.
iv. The PFL shall periodically assess that the information and security standards are met by the cloud which includes information security, privacy, data residency, usage, retention and disposition, repatriation, availability business continuity planning, emergency response and accessibility during the period of validity of the contract and also in the event of a planned or unplanned termination of the cloud services.
v. PFL shall ensure that data/ information residing on the cloud can be replicated/ backed-up and be accessible to the PFL in the event the Cloud becomes in accessible (e.g due to a disaster event or termination of service) as per the PFL's business continuity policy.

34. Applications that can be hosted on public cloud or private cloud or on-premise infrastructure
35. PFL shall conduct periodic feasibility assessment of the services provided and the applications hosted on the cloud in order to decide to roll-back or shift to other public cloud or the PFL's private cloud, as deemed appropriate. PFL shall leverage appropriate solutions (such as VMware) while using public/ hybrid/ multi cloud (as appropriate) to seamlessly move back to private cloud.
36. PFL shall consider using hybrid/ multi cloud for applications that require varying computational capability (even for limited period) and unpredictable load based on the security compliance as per the Information Security Policy and the cost benefit.
37. PFL shall conduct a *Cloud Suitability Assessment"" exercise for a given business need to select the most suitable option/ combination of public/ private/ hybrid/ multi cloud or on-premises infrastructure, and the delivery options of IaaS, PaaS or SaaS. Competent authority in IT shall approve the outcome of the "Cloud Suitability Assessment" exercise.
38. PFL shall use the cloud to fulfil the business objectives, minimize the risks and minimize the duplicate services or infrastructure in the PFL
39. To optimally and efficiently use the cloud, PFL shall create a suitable cloud governance structure to empanel the cloud service providers, determine the hosting model, monitor & manage the cloud, and ensure compliance to the information security and regulatory requirements.
40. PFL shall maintain an inventory of all empanelled Cloud Service Providers along with the contract/ SLAs that includes performance, availability and support expectations.
41. PFL shall actively monitor the market trends/ disruptions and adjust the empanelment of cloud service provider.

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42. The PFL shall use private cloud, after detailed assessment, to host applications that require dedicated infrastructure, high performance access to file systems, heavily trafficked applications.
43. Ease of Consumption: PFL shall maintain and regularly refresh a catalogue of pre-approved service offerings of the cloud including directly consumable self-service options.
44. The cloud shall also provide the PFL with the right, opportunity, means of finding, viewing, using, and retrieving data/ information to the authorized users.
45. Agility: PFL shall ensure that the cloud eliminates/ reduces the delays in procurement and commissioning of infrastructure, software and related service, and provide agility to the PFL. Certain applications (such as HR Analytics) can be consumed as SaaS from the cloud.
46. Scalability & Rapid Elasticity: PFL shall ensure that the cloud provides on-demand capacity and allows scale-up/ scale-out dynamically for sudden business/ technical needs.
47. High Availability: PFL shall ensure that the cloud provides the desired availability as per the application and business requirements and information security (as per the Information Security Policy)
48. Geographical Coverage & Compliance: PFL shall ensure that the cloud complies with data sovereignty, data privacy, compliance guidelines of the PFL and the regulatory / statutory requirements catering to the distributed nature of users or applications.
49. Metering: PFL shall ensure that the cloud provides a transparent mechanism to monitor, control and report the usage of the resources using he" a er metering of model as per the PFL's requirements,
50. Security & Privacy: PFL shall ensure that data/ information in the cloud is protected.

    from unauthorized access, use, disclosure, disruption, modification, inspection, recording, or destruction, and complies with the Information Security Policy of the PFL Guidelines for On-Premise & Private Cloud
51. PFL shall regularly monitor the capacity needs for cost optimization.
52. Any exception to this policy should be approved by the CIO / CTO (CTO means Chief Technology Officer and / or Chief Transformation Officer).